Ember: the long life of a $0.1M_\odot$ star

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Ember is a one-dimensional stellar evolution code that calculates a star's structure, composition and energy transport as it ages. We use it to follow a star of $0.1M_\odot$ through 3.875×10^{12} yr from a static model at the beginning of hydrogen burning. Its central hydrogen mass fraction has fallen to $X = 0.001294$, while the surface remains hydrogen-rich at $X = 0.1730$. A core that transports heat by radiation and conduction contains 95.44% of its mass, surrounded by a convective envelope. Hydrogen burning still supplies almost all the luminosity. The aim is to calculate the transition to a helium white dwarf, cooling through 1000, 500 and 100 K, and subsequent evolution through disappearance under explicit nucleon-decay scenarios. Section 8 gives the lifetime timeline and its conditional alternatives.

The figures show the evolutionary track through the model at the atmosphere table's 5000 K surface temperature limit. We compare this trajectory with the published $0.1M_\odot$ model of Laughlin, Bodenheimer and Adams (1997; LBA97), using their stated values and approximate curves read from their figures. To make a direct numerical comparison, we have generated an f77-source code that seeks to replicate LBA97's exact assumptions and methods.

1 The star and its evolution so far

The initial hydrogen mass fraction is $X_{\mathrm{H}} = 0.7$, the initial ^{3}He fraction is zero, and the metal mass fraction is $Z = 0.02$. The metals follow the relative elemental abundances of Grevesse and Sauval's GS98 mixture, with the remaining initial mass in ^{4}He. The calculation holds baryonic mass fixed and omits rotation, magnetism, mass loss and external heating. Its age excludes formation and pre-main-sequence contraction.

Model at 3.875Tyr	512 points
Radius ($R_\odot$)	0.1241
Photon luminosity ($L_\odot$)	0.008674
Effective surface temperature (K)	5000
Central temperature (MK)	12.61
Central density (g cm^{-3})	6757
Central hydrogen mass fraction	0.001294
Surface hydrogen mass fraction	0.1730
Central ^{3}He mass fraction	1.862×10^{-9}
Remaining hydrogen mass ($M_\odot$)	0.007414
Convective mass fraction	0.04563

The plotted sequence contains the initial stellar model and 8455 successive models after accepted time steps, each resolved into 512 mass points. Each time step evolves the abundance profile and solves the stellar structure and energy balance, including energy released or absorbed as the structure changes. The calculation uses 196 gas atmospheres, 72 equation-of-state composition tables, and Los Alamos ATOMIC radiative opacities for the high-temperature interior [1, 2]. The latter include additional hydrogen-poor mixtures through $X = 0$, used above $T = 5.012 \times 10^5$ K; the lower-temperature opacity tables retain their separate composition limits.

1.1 Comparison of the starting model with observed stars

The starting radius, luminosity and temperature can be compared with nearby low-mass stars and an eclipsing companion very close to $0.1M_\odot$. The following values are reported by Davis et al. [3], Agol et al. [4], and Ribas et al. [5].

Star	Mass ($M_\odot$)	Radius ($R_\odot$)	T_{eff} (K)	Luminosity ($L_\odot$)
Ember, initial	0.1	0.1259	2766	0.0008362
EBLM J2114–39 B	0.0993 ± 0.0033	0.1250 ± 0.0016	—	—
TRAPPIST-1	0.0898 ± 0.0023	0.1192 ± 0.0013	2566 ± 26	0.000553 ± 0.000019
Proxima	0.120 ± 0.003	0.146 ± 0.007	2980 ± 80	0.00151 ± 0.00008

The strongest comparison is EBLM J2114–39 B: its mass agrees with $0.1M_\odot$, and Ember's radius is only 0.8% larger, within the reported radius uncertainty. The binary's nearly solar metallicity, [Fe/H] $= -0.001 \pm 0.032$, also makes it relevant to our adopted composition. Its mass and radius are inferred from eclipses and orbital velocities together with an estimate of the primary's radius; its temperature and luminosity are not measured in that study.

Ember falls between the lower-mass TRAPPIST-1 and higher-mass Proxima in radius, temperature and luminosity. These comparisons are less direct: TRAPPIST-1's adopted mass uses an empirical mass–luminosity relation, and the quoted Proxima mass comes from stellar models. Age, composition and magnetic activity also differ between individual stars. Within Ember, roughly ten billion years of hydrogen burning changes luminosity by less than 0.4% and temperature by less than 2 K, so ordinary main-sequence age does not materially change this comparison. Formation and early contraction are excluded. Together, the radius agreement at nearly the same mass and the plausible temperature and luminosity support using this initial stellar structure for the calculation. They do not yet establish percent-level accuracy or select uniquely among the atmospheric and interior physics assumptions.

2 Comparison with the published LBA97 model

The main comparison is with Laughlin, Bodenheimer and Adams [6], here abbreviated LBA97. Their section 3.2 and Figure 1 describe the full hydrogen-burning evolution of a $0.1M_\odot$ star and its entry into white-dwarf cooling. We use numbers stated in the text and approximate values read from the published figure.

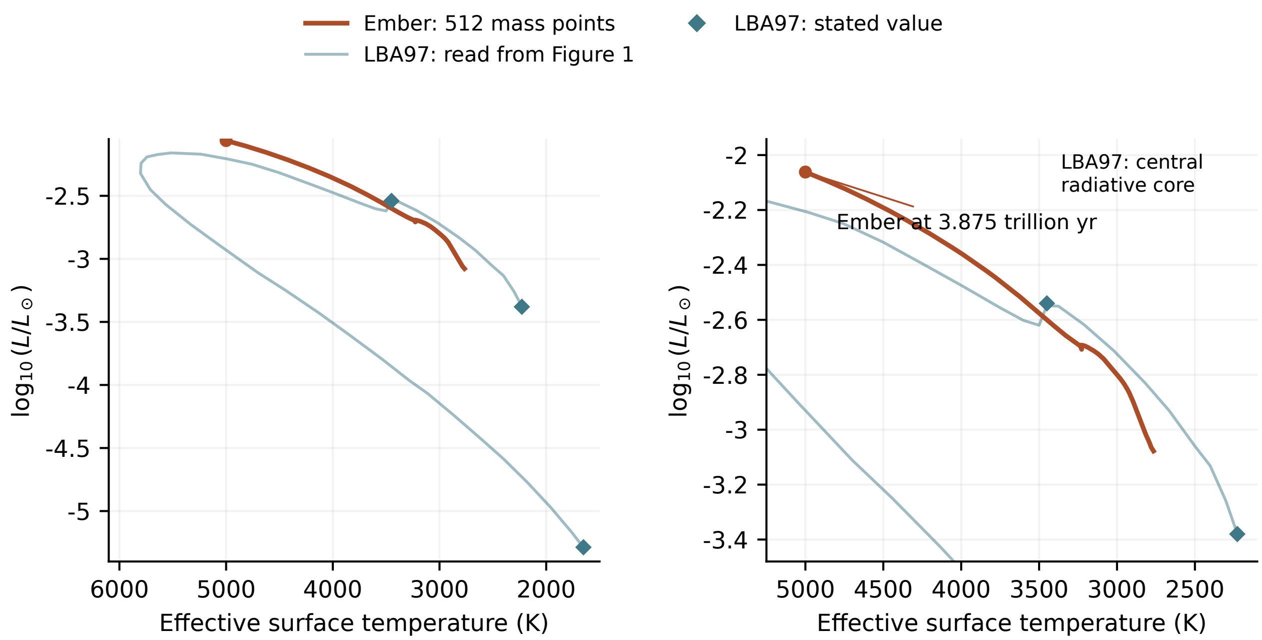

Figure 1: Ember compared with the published LBA97 evolution. The pale curve was read from Figure 1 of LBA97; diamonds mark values stated in their text. The left panel shows the full published path from the start of hydrogen burning through cooling. The right panel enlarges the region reached by Ember. The orange curve contains the complete 512-point Ember calculation through 3.875Tyr. The original pre-main-sequence contraction and the dotted diagonal guide in LBA97 are excluded. The pale line connects 42 selected positions on the printed track; it is an approximate reconstruction, not the original numerical output. Neither temperature nor luminosity has been adjusted to make the curves agree.

Quantity	LBA97, published	Ember
Initial surface temperature (K)	2228	2766
Initial luminosity ($L_\odot$)	$\simeq$ 0.000417	0.0008362
Initial central density ($\mathrm{g\,cm^{-3}}$)	309	381.7
Maximum ^{3}He mass fraction	0.0995	0.1003
Age at maximum ^{3}He (Tyr)	1.38	0.7103
Central radiative core	Forms at 5.742Tyr	Present at 3.875Tyr
Hydrogen mass fraction at core onset	$\simeq$ 0.16	Not yet measured
Highest surface temperature (K)	5807	Not yet reached

The two starting states are defined differently. LBA97 includes contraction and defines main-sequence arrival by nuclear reactions supplying all surface luminosity, after about 2 billion years. Ember starts from a static hydrogen-burning model at age zero. That clock offset is small compared with the later differences, but the initial thermal and atmospheric models also differ. Ember initially radiates about twice the published luminosity and has a hotter surface. These comparisons should not be interpreted as a measurement of the error in one isolated physical ingredient.

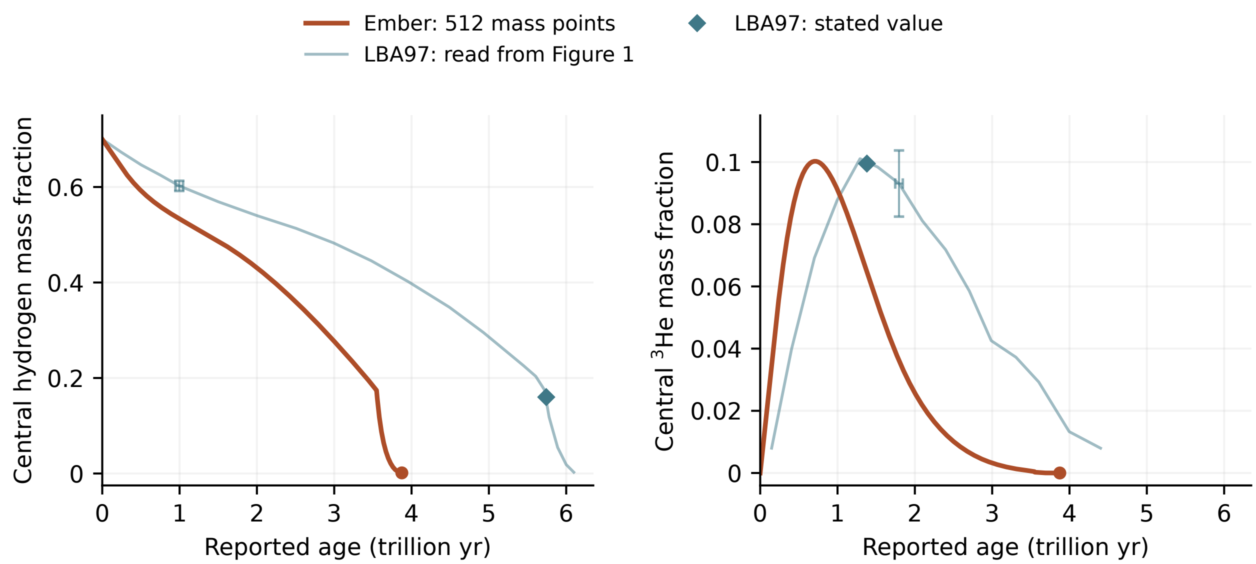

Figure 2: Hydrogen consumption and the buildup and depletion of ^{3}He. The pale curves are read from the inset of LBA97 Figure 1, with 20 hydrogen and 15 helium-3 samples. Diamonds show their stated ^{3}He maximum and the hydrogen abundance at central radiative-core formation. Orange curves are the 512-point Ember track through 3.875Tyr. Ages are shown as reported, without shifting or stretching either calculation. The representative crosses correspond to four image pixels: about 0.05Tyr in age and 0.01 in mass fraction. They illustrate reading precision, not statistical or physical uncertainty. The rapid final hydrogen decline is poorly resolved in the printed inset; the helium-3 trace ends as it becomes indistinguishable from the baseline.

The maximum ^{3}He abundance agrees to within about 1%, but Ember reaches it in about half the time. Ember also consumes hydrogen faster over the interval calculated so far. Reading the published hydrogen curve gives an age of roughly 5.6Tyr at hydrogen fraction 0.2034, reached by Ember near 3.40Tyr. The model at 3.875Tyr has central hydrogen below 0.16, the approximate abundance stated by LBA97 at its central-core transition. The figure does not justify a more precise age comparison. The luminosity–temperature track alone can conceal this difference because it does not show how long the star spends at each position.

LBA97 used a grey atmosphere, approximate opacity changes as hydrogen is consumed, and older interior opacities and nuclear rates. Ember uses frequency-dependent gas atmospheres, composition-specific opacity tables, a different equation of state and updated pp-chain rates. Grain formation is still omitted from Ember's evolving atmosphere. These differences are candidates for the changed surface properties and hydrogen-consumption rate; comparisons that change one ingredient at a time are needed to separate them. Agreement with LBA97 and computing time alone do not establish physical accuracy.

2.1 Core transport by radiation and conduction

The track first departs from full convection near 3.546Tyr, at central hydrogen fraction $X = 0.1751$. By 3.560Tyr, the convective mass fraction has fallen to 0.6277 and central hydrogen to $X = 0.1518$, while surface hydrogen remains at $X = 0.1747$. This separation of central and surface composition shows that convection no longer mixes the whole star.

At 3.875Tyr, the model has a core containing 95.44% of the mass that transports heat by radiation and conduction, surrounded by a convective envelope. Central hydrogen has fallen to $X = 0.001294$, while the envelope remains near $X = 0.1730$. Radiation and conduction can carry the required energy through this core without convection; the composition gradient also suppresses convection. Conduction supplies 84.71% of the local heat flux at the center, with radiation carrying the remaining 15.29%. Conduction is therefore already significant in the central region. Figure 3 shows the declining convective mass and the final abundance profile on this track. The first loss of full convection does not by itself establish a central radiative core. The final profile establishes the core but does not determine its formation age. LBA97 reports central radiative-core formation at 5.742Tyr.

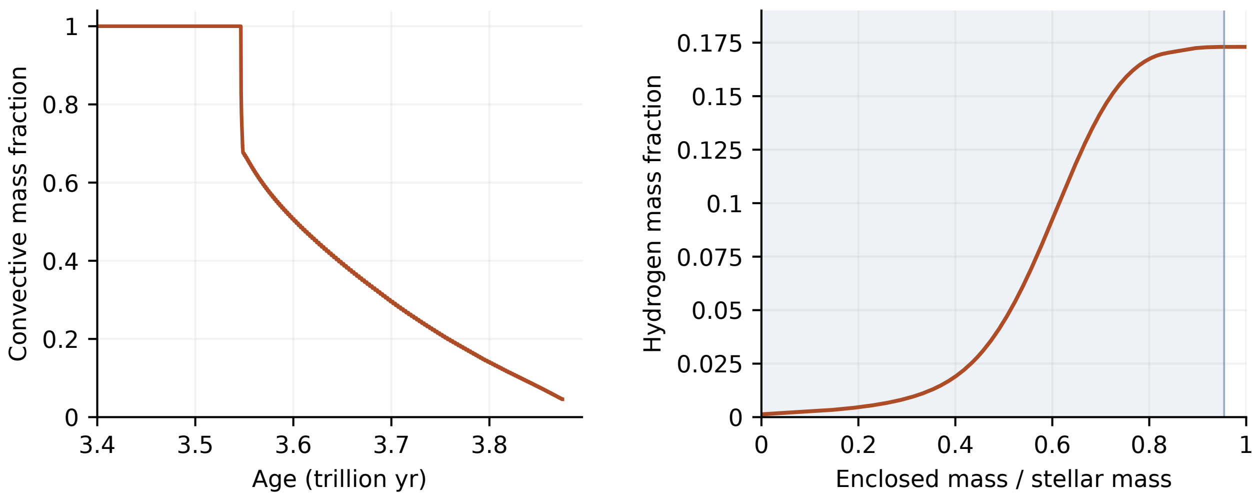

Figure 3: Convection and chemical separation along the Ember track. Left: the fraction of mass in convective regions declines from unity near 3.546Tyr to 0.04563 at 3.875Tyr. Right: the hydrogen profile in the 3.875Tyr model. The shaded core carries heat by radiation and conduction; the outer convective envelope has an almost uniform hydrogen abundance. Both panels use the same calculation and its final saved structure.

"Fully convective" describes where convection occurs, not the fraction of the luminosity it carries. Radiation, conduction and convection share the heat flow continuously. The model's mixing-length calculation makes the convective contribution approach zero at a stability boundary. Chemical mixing uses a stronger approximation: each connected convective region becomes uniform rapidly. Slow composition mixing is included in some marginally stable regions, but their additional heat transport still needs attention.

2.2 The reconstructed F77 program is a separate comparison

The Henyey/F77 program is useful for controlled code comparisons, but it does not reproduce the LBA97 calculation. Its AJR-opacity run first flags the center as nonconvective at 3.224Tyr and $X_{\mathrm{H}} = 0.01756$; its Ferguson-opacity run does so at 2.946Tyr and $X_{\mathrm{H}} = 1.064 \times 10^{-5}$. Both differ substantially from the published 5.742Tyr and $X_{\mathrm{H}} \simeq 0.16$. The AJR run's highest surface temperature, 5936 K, is much closer to LBA97's 5807 K, but agreement in that one quantity does not validate the full history.

The F77 reconstruction substitutes some opacity and nuclear-rate prescriptions. At $X_{\mathrm{H}} = 0.2034$, it is about 76% more luminous than Ember; switching only its low-temperature opacity choice changes its luminosity by 54.5%. These comparisons establish sensitivity to the inputs, rather than identifying the source of the Ember–LBA97 difference. The separate comparison notes and supplementary F77 graphs retain these results. The F77 curves are thin and translucent, with abrupt changes drawn more faintly and all accepted data preserved.

2.3 Photospheric evolution over the opacity table

Figure 4 follows the presentation of LBA97 Figure 6: photospheric density and temperature over an opacity map, with circle diameter proportional to stellar radius. For Ember, we recover the photospheric state at Rosseland optical depth 2/3 from the accepted atmosphere structures. The local temperature there differs from the effective temperature of a frequency-dependent atmosphere. It rises from 3019 to 5166 K over the plotted history, while density falls overall from 1.432×10^{-5} to $3.203 \times 10^{-6}\ \mathrm{g\,cm^{-3}}$. The two opacity maps use the same axes and color scale. The LBA97 map is inferred from the published shading, calibrated against the tabulated opacities of Alexander, Johnson and Rypma [7]. Nineteen cells excluded from the calibration differ by at most 0.12 in $\log_{10}\kappa$; this checks the approximate scale of the diagram, not the accuracy of its grain physics.

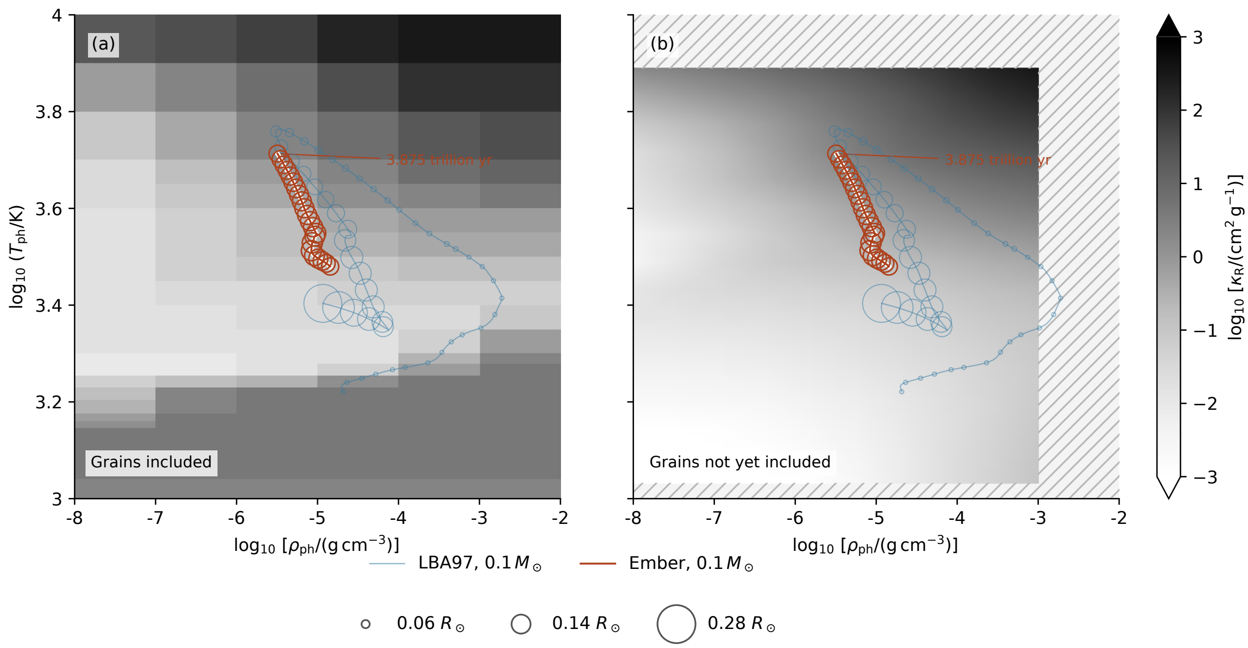

Figure 4: Photospheric evolution of a $0.1M_\odot$ star over hydrogen-rich opacity maps: (a) the grain-inclusive LBA97 Figure 6 map on an approximate calibrated opacity scale; (b) the Rosseland absorption mean of Ember's SYNSPEC wavelength table at $X = 0.7$, $Z = 0.02$ and zero ^{3}He. Both panels use identical axes and grayscale normalization. The right map excludes scattering and grains; the low-temperature grain-opacity band in the left map is therefore absent. Hatching marks missing wavelength-table coverage. Both panels show the LBA97 track recovered from 897 figure markers and Ember's calculated track through 3.875Tyr. The LBA97 curve includes contraction and cooling; the plotted Ember models are burning hydrogen. Open-circle diameters scale with radius, with 48 LBA97 and 22 Ember samples. LBA97 radii are normalized using its stated main-sequence luminosity and temperature. Ember's local photospheric temperature and density are interpolated at Rosseland optical depth 2/3 using the evolving surface composition. The fixed backgrounds do not represent the changing opacity of either star.

Photospheric temperature and density are interpolated in surface composition, effective temperature and gravity using the atmosphere's molecular equation of state. The corresponding interpolation at the deeper stellar matching boundary reproduces its temperature and pressure to numerical precision at 65 sampled models.

3 Physical ingredients in Ember

Component	Treatment and present limits
Equation of state	FreeEOS 3.0 supplies pressure, internal energy and their responses to temperature, density and composition. Electron integrals are evaluated numerically. Interpolation is based on a shared Helmholtz free energy to keep these responses consistent. The selected 72 composition tables include zero hydrogen and the low-density states needed at the atmosphere boundary. They do not describe the cold, dense remnant.
Opacity	AESOPUS 2.1 provides low-temperature gas opacity, and the LANL TOPS service supplies ATOMIC opacities for hotter and denser conditions [1, 2]. The two are joined only where their available ranges permit it. Hydrogen-poor compositions have been refined and checked against additional calculations. The ATOMIC tables reach zero hydrogen; the cool opacity still requires hydrogen-rich mixtures.
Hydrogen burning	A reduced ppI/ppII network follows ^{3}He out of equilibrium. It uses Solar Fusion II reaction data and finite-degeneracy Salpeter–Van Horn screening, which accounts for the plasma's effect on charged reactants. Atomic mass differences determine heat and nuclear-neutrino energy. Pep, hep, ppIII and CNO reactions are absent; metals do not react.
Heat and composition transport	Mixing-length convection uses the Ledoux criterion, including the stabilizing effect of composition gradients. Slow mixing associated with semiconvection and thermohaline instability is included. Conduction uses approximate mixtures of Ioffe/Cassisi inputs. Microscopic diffusion, settling and a separate semiconvective heat flux are absent.
Atmosphere	TLUSTY208/SYNSPEC54 calculates plane-parallel gas atmospheres in local thermodynamic equilibrium, resolving absorption and scattering over frequency. The interior joins the atmosphere at optical depth 100. The atmospheres include atomic and molecular absorption and collision-induced absorption, but no accepted condensate treatment.
Thermal neutrinos	Plasma-decay losses follow Haft, Raffelt and Weiss [10] and enter the energy equation with analytic derivatives. Other thermal-neutrino processes are not included in this choice.
Numerical evolution	Coupled implicit equations solve structure, burning and mixing. Comparing a full time step with two half steps controls time accuracy. The mass mesh is fixed. Saved models include enough internal information to continue with the same executable, tables and numerical settings.

The interior equation of state, atmosphere chemistry, opacity and conduction come from different physical sources. Their agreement therefore needs checking; accurate derivatives within one component do not prove that all components describe the same thermodynamics. FreeEOS omits potassium from the

GS98 mixture, a missing mass fraction of 4.56×10^{-6}. Helium-isotope and atomic partition-function approximations also remain.

With all other inputs fixed, replacing the Solar Fusion II rates with Solar Fusion III [14] raises the initial luminosity by 0.7454% and radius by 0.2164%. Both rate choices have been evolved through 10^{10} yr for this comparison. The plotted trajectory uses Solar Fusion II; the short comparison does not determine a change in the full hydrogen-burning lifetime.

4 Why more atmosphere calculations are needed

The atmosphere calculations are reusable. They are performed separately and stored as tables of boundary temperature and pressure. During stellar evolution, Ember interpolates those tables; it does not solve frequency-dependent radiative transfer at every time step. New atmosphere calculations are needed when the star reaches a composition, surface temperature or gravity not adequately covered, or when the physical treatment changes.

The plotted track uses 196 gas atmospheres, spanning the combinations of surface temperature, gravity and composition that it visits through 5000 K. The tabulated source states include hydrogen fractions from 0.1 to 0.7 and ^{3}He fractions from zero to 0.12. These outer ranges do not imply that every combination has been calculated: interpolation requires a complete set of surrounding source models, including those needed for derivatives.

Independent atmosphere calculations test the interpolation. At two 3700 K models with compositions between the tabulated values, matching temperatures differ by −0.4837% and −0.7399%, and gas pressures by +0.5707% and +0.5137%. Changing the lower boundary of the source atmospheres changes matching values by at most 0.007883% in the specified comparisons. All 196 source models have support in the selected 72-table equation of state. Independent comparisons with numerical-electron FreeEOS at 4736 states give a maximum heat-capacity difference of 0.2952%; 2240 states also pass thermodynamic identity checks. At the final stellar model, independent FreeEOS evaluations at all 512 actual compositions, temperatures and densities differ by at most 0.06584% in heat capacity. Interpolation uses thermodynamically consistent source states and excludes unsupported derivative stencils. These are sampled comparisons, not error bounds across the full table.

At an independently calculated 3900 K atmosphere, interpolated matching temperature differs by −0.4427% and gas pressure by +0.4107%. The largest change in four lower-boundary comparisons is 0.005127%. Independent atmospheres at 4100 and 4300 K differ from the table by at most 0.5986% in the matching state; eight lower-boundary comparisons change it by at most 0.006377%. At 4500, 4700 and 4900 K, independent matching temperatures differ by at most 0.05225% and gas pressures by 0.5376%. Twelve lower-boundary comparisons at 4600–5000 K change matching values by at most 0.007810%. Equilibrium chemistry finds no condensates in the tested structures from 3800 through 5000 K.

The high-temperature ATOMIC opacity tables include additional mixtures with $X = 0$, 0.025, 0.05 and 0.075 at three metal fractions. Twelve independent composition comparisons on hot stellar-profile states differ by at most 0.05257%. This extends the composition coverage needed as central hydrogen is depleted; the lower-temperature tables have their own composition limits.

4.1 Condensates and a hydrogen-free atmosphere

The current evolving atmosphere uses gas-only opacity. Independent chemistry checks found no condensates in the tested 3000–3200 K atmospheres, but found them in the cool upper layers of 2600–2800 K models. Effective temperature alone is therefore insufficient to decide whether grains form. Checks on the warm hydrogen-poor structures through 5000 K find no condensates; colder structures still require atmosphere calculations with grain formation and opacity included.

Grain formation can affect radiation in two ways: it removes absorbing atoms and molecules from the gas, and any suspended grains absorb and scatter light. A calculation that isolates gas depletion assumes the condensed material settles out. At solar composition and 2800 K, gas depletion changes the interior matching pressure by 0.1246% and temperature by −0.02347%. These are local gas-depletion effects, not measurements of grain opacity or of a stellar lifetime change. Dusty atmosphere models become relevant around 2800 K and below [11]. A separate grain-opacity calculation uses published optical data and explicit grain-size choices. It passes comparisons with an independent scattering program and analytic small-particle limits. Applied to a fixed gas atmosphere, it combines the optical properties with the computed alumina abundance. The maximum condensed mass fraction is 1.094×10^{-4}. For compact grains of radius $0.1\,\mu$m, the integrated grain optical depths near $1\,\mu$m are 0.002610 for absorption and 0.002829 for scattering. These values describe one fixed 2800 K gas profile with suspended grains, not a recomputed atmosphere or a prediction of grain size. Absorption and scattering remain separate. Coupling these opacities to the atmospheric structure requires consistent material phases, wavelength coverage and thermal response, as described in the grain-opacity notes.

Grain formation also changes thermodynamics through redistribution of atoms among molecules, ions and solids. Comparing gas-only and gas-plus-grain chemical equilibria at one test state gives a heat-capacity correction of $1.175 \times 10^5\ \mathrm{erg\,g^{-1}\,K^{-1}}$, compared with 2.890×10^5 from counting grains alone. A consistent thermodynamic reference, abundance normalization and treatment across phase boundaries are required before this contribution can enter the evolving model.

The atmosphere wavelength-opacity tables cover material temperatures through 1.315×10^4 K. Nonfinite absorption near He I 4471 Å prevents use of the present source at 1.500×10^4 K. Atmosphere calculations must remain within the valid opacity domain, with matching conditions insensitive to the chosen lower boundary.

A hydrogen-free atmosphere presents a further issue. The source programs' abundance normalization and molecular initialization currently depend on hydrogen. A direct opacity test at 4017 K shows that multiplying all number abundances by the same factor preserves composition and nearly preserves electron density, but can change individual absorption coefficients by a factor of 27.3. This one-temperature test does not measure a stellar-boundary error; it shows that a simple abundance rescaling is not yet a verified solution. The hydrogen-free limit therefore requires a source formulation that does not normalize abundances to hydrogen.

5 Energy and the cold helium remnant

The discrete energy equation used for stellar evolution is

$$L_\gamma = L_{\mathrm{nuc,dep}} + L_{\mathrm{grav}} - L_{\nu,\mathrm{thermal}}, \qquad L_{\mathrm{nuc,rest}} = L_{\mathrm{nuc,dep}} + L_{\nu,\mathrm{nuclear}}.$$

Here L_γ is the photon luminosity, $L_{\mathrm{nuc,dep}}$ is nuclear heat left in the star, and L_{grav} is the contribution from changing internal and gravitational energy. Nuclear and thermal neutrinos are counted separately.

At 3.875Tyr in the 512-point model, in solar luminosity units,

$$L_{\mathrm{nuc,dep}} = 0.008653, \quad L_{\mathrm{grav}} = 2.072 \times 10^{-5}, \quad L_{\nu,\mathrm{nuclear}} = 2.566 \times 10^{-4}, \quad L_{\nu,\mathrm{plasma}} = 4.861 \times 10^{-8}.$$

The changing-energy term refers to the last implicit half step; the other reported diagnostics use the saved state. Before rounding, the discrete fractional luminosity residual at this saved endpoint is below 4×10^{-11}. This checks the equations implemented in the code, not an independent exact energy integral or the absence of missing physical processes.

Cold degenerate electrons have a small thermal response superimposed on a large temperature-independent contribution. Paired numerical integration around the Fermi surface retains the small response without subtracting nearly equal large terms. Pure-helium tests at 100–10^4 K and densities 10^3–10^6 $\mathrm{g\,cm^{-3}}$ agree with the expected low-temperature limit to about 10^{-10} fractionally at 100 K. Electron entropy and ionic translational entropy, including mixing, are also implemented and checked against thermodynamic identities. These analytic components have not yet been combined into the production equation of state for a remnant.

Liquid and solid free energies and their derivatives are compared with Ioffe EOSFI22. Across sixteen forced-phase tests, the largest checked density-response discrepancy is 1.05×10^{-6} in ideal-ion pressure units. This tests derivative consistency within each phase; phase selection requires comparing the free energies themselves.

Helium quantum melting, mixture effects, latent heat and the connection to partially ionized outer layers remain unfinished. The source's default classical melting value, $\Gamma = 175$, cannot stand in for a helium phase calculation [12]. Dense, cool atmospheres also require nonideal chemistry and pressure-dependent absorption, including collision-induced absorption; refraction and dense-fluid effects need checking [13].

6 Hydrogen exhaustion and remnant cooling

Extending the track through hydrogen exhaustion requires atmosphere and interior inputs covering the changing composition, temperature and density, including hydrogen-free mixtures and grain formation where relevant.

As composition becomes stratified, conservative microscopic diffusion and settling must include the electric field that maintains charge balance and the associated energy exchange. Moving convection boundaries and thin burning layers need enough mass resolution; an adjustable mesh is required if the fixed mesh does not converge. The importance of omitted nuclear reactions, thermal-neutrino processes and semiconvective heat transport must be assessed in the conditions actually reached.

Hydrogen exhaustion and cooling require distinct measurements. We will record central hydrogen fractions crossing 10^{-3}, 10^{-4} and 10^{-5}, retaining the accepted models on either side. We will also record when nuclear heat falls below 10%, 1% and 0.1% of photon luminosity during cooling, including any later recovery. Remaining hydrogen mass and shell burning must be followed; a depleted center alone does not establish that burning has ended everywhere.

A consistent liquid/solid helium equation of state, conductive transport and a dense atmosphere must allow the star to evolve through 1000, 500 and 100 K on the descending-temperature branch. These endpoints must converge as mass resolution, time accuracy and physical tables are refined. The resulting cooling ages depend on the thermal reservoir, phase changes and atmospheric boundary as well as the numerical accuracy.

The present calculation assumes fixed mass and no external heating. External heating, stellar encounters and particle decay are treated as conditional extensions of the isolated track [8]. The timeline in Section 8 includes these alternatives, cooling past 100 K, loss of degeneracy during mass depletion, and the end of the bound remnant.

7 Computing time and reproducibility

The plotted 512-point calculation, from the initial model to the atmosphere limit at 3.875Tyr, used 120.7 CPU minutes and 99.27 elapsed minutes, with 8455 accepted steps and 319 rejected attempts. These timings were measured while other calculations shared the processor. Atmosphere generation is excluded from the stellar CPU time. The calculation reuses exactly matching nuclear responses and shares electron calculations within each response. This is a measured partial trajectory; the cost of hydrogen exhaustion and cooling remains unmeasured.

Repeated nuclear calculations are an identified production cost. Enlarging the cache of exactly matching stellar states reduced CPU time by 42.21% in a short alternating comparison. Sharing the electron calculation within each nuclear response saved a further 14.97% in a separate short comparison with the enlarged cache. All full output files in each comparison agree exactly; 2048 saved late stellar states also give identical nuclear responses and derivatives with the additional electron reuse. Both forms of reuse are included in Ember. A fixed-input timing comparison through 3.757Tyr isolates the larger cache with identical physical inputs and entire output files: CPU time falls from 162.4 to 101.8 minutes, a reduction of 37.31%. The additional electron reuse was not part of that pair. Concurrent background load varied, so this single long pair is not an isolated-machine timing guarantee. The savings during shell burning and remnant cooling remain unmeasured.

The unmodified F77 AJR run used 27.37 CPU seconds through its programmed cooling stop. Ember's finer mass mesh, coupled implicit solves, time-step checks and different physics perform additional work. Their separate contributions to the full speed difference have not been measured. An atmosphere replay spent about 54% of its measured time in chemistry and only 0.0033 seconds in the dense matrix solve; that describes one atmosphere calculation, not the stellar integrator.

The GitHub repository contains code, table-generation instructions, small records of input choices and checks, this paper and the numerical data needed for its figures. The LBA97 comparison records the selected image coordinates, axis calibration and original PDF checksum; its curves can be rebuilt without the original image. The F77 figure data are extracted from accepted models and checked against the recorded output hashes. Full raw outputs, generated physical tables, saved stellar states and executable archives remain local. Instructions for recreating the figures and recovering the recorded stellar calculation accompany the paper files.

Ember passes all 34 test suites with the installed physical tables. The repository supplies figure data, source specifications and validation records; regenerating the bulk physical tables is required for stellar calculations from a fresh checkout.

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8 Lifetime timeline

The timeline combines the calculated hydrogen-burning track with the later physical stages to be evaluated. Numerical ages labelled Ember come from the track plotted in this paper. Formation precedes its age zero; subsequent cooling ages have not yet been calculated. Environmental events and material transitions can overlap, so their order is conditional. Disappearance means the loss of the baryonic remnant, with energy carried away by decay products.

8.1 Formation and hydrogen burning

Age or condition	Event	Physical description and basis
Before $t = 0$	Formation and accretion	Cloud collapse assembles the protostar. Accretion, a disk and any outflows set its initial mass, entropy and angular momentum. These stages are outside the present fixed-mass calculation.
Before $t = 0$	Deuterium burning and contraction	The young object contracts toward sustained hydrogen burning; deuterium is consumed, and lithium is depleted as the interior heats. Low-mass pre-main-sequence models supply the comparison, rather than an Ember formation age [15].
$t = 0$	Fully convective hydrogen-burning model	Ember begins with $X = 0.7$, $Z = 0.02$, $R = 0.1259R_\odot$ and $T_{\rm eff} = 2766$ K. The pp chain converts hydrogen to helium while convection mixes the star.
0.7103Tyr	Maximum ^{3}He abundance	The calculated central ^{3}He mass fraction reaches 0.1003 and then decreases as its destruction catches up with its production. LBA97 gives 0.0995 at 1.38Tyr.
3.546Tyr	Loss of full convection	The first saved Ember structure with a nonconvective region has central $X = 0.1751$. This does not alone date formation of a central radiative core. LBA97 reports its central radiative core at 5.742Tyr and $X = 0.16$.
3.560Tyr	Separate core and envelope composition	The calculated convective mass fraction is 0.6277. Central $X = 0.1518$ differs from surface $X = 0.1747$; whole-star mixing has ended.
3.875Tyr	Endpoint of the plotted track	$T_{\rm eff} = 5000$ K and central $X = 0.001294$. Conduction and radiation carry heat through the core; the convective envelope contains 4.563% of the mass. Hydrogen burning still dominates luminosity.
After the plotted endpoint	Central exhaustion and shell burning	Record crossings of central $X = 10^{-3}$ and 10^{-4}, the remaining hydrogen mass, and burning outside the depleted center. Central exhaustion does not imply exhaustion of the envelope.
Age to be calculated	Turn toward white-dwarf cooling	Track the maximum surface temperature and the change to sustained cooling. LBA97 reaches 5807 K; this is a comparison value, not an imposed Ember turn. The low mass leads toward a helium remnant without normal core helium burning.

8.2 Cooling and environmental exposure

These stages are identified by local conditions or their effect on the energy balance. A surface temperature of 100 K is an intermediate milestone; it does not imply a 100 K core. The calculation will compare the following processes before assigning their event ages.

Condition	Event	Physical description and calculation
Declining nuclear power	Residual hydrogen burning fades	Record nuclear heating below 10%, 1% and 0.1% of photon luminosity, including any later recovery. Retain the envelope fuel supply and thermal energy rather than assuming an immediately passive remnant.
Transport competes with evolution	Diffusion and separation	Hydrogen can collect at the surface while heavier species settle. The atmosphere's composition follows transport and mixing; a helium core does not imply a helium atmosphere.
Envelope cools	Molecules, grains and dense atmospheres	Follow molecular absorption, collision-induced absorption, scattering, condensation and settling where the local conditions permit them. Couple opacity to composition and energy transport [11, 13].
Interior reaches a phase boundary	Crystallization and chemical separation	Evaluate helium and mixture phase equilibria, latent heat, separation energy and solid heat capacity. The front's progress is determined by density and core temperature, not one assigned surface temperature [12].
Convection reaches deeper layers	Envelope and core become thermally coupled	Determine whether and when the convective envelope reaches the nearly isothermal interior, changing the relation between stored heat and luminosity. Its order relative to crystallization is to be calculated.
$T_{\text{eff}} = 1000$, 500, 100 K	Cold-remnant milestones	Record ages, core temperature, radius, composition, solid fraction and power sources on the cooling branch. Below 100 K, continue with the appropriate solid and partly neutral matter description.
Including ages $\gtrsim 10^{14}$ yr	Changing environment and accretion	Compare retained and ejected objects in evolving stellar and dark-matter environments. Gas, debris and interstellar solid objects provide conditional accretion and heating; no unique late encounter rate is assumed.
Close passages	Tidal heating, capture or collision	For surviving unbound WD pairs, distinguish heat from spin and persistent oscillations. Couple an encounter survey to waiting-time distributions and cooling after each pulse; captures and collisions form separate outcomes.
Allowed particle interactions	Dark-matter heating	Compare capture, thermalization, evaporation and possible annihilation with the photon luminosity. Include no annihilation heating as an alternative; crystallization can change thermalization [20].
Nuclear times become competitive	Density-driven fusion	Estimate pycnuclear and impurity reactions against cooling and decay times, using the actual helium-rich composition. Carbon-burning results do not directly determine helium-remnant rates [21].

8.3 Decay and disappearance: conditional branches

Proton decay has not been detected. The Super-Kamiokande limit for $p \to e^+\pi^0$ is a partial lifetime $\tau_p/B > 2.4 \times 10^{34}$ yr at 90% confidence [16]. Baryogenesis does not itself require ordinary proton decay; leptogenesis provides a counterexample [17].

Virtual-black-hole interactions supply another possible decay channel. A Planck-scale estimate is $\tau_p \sim 10^{45}$ yr, with rate set by the assumed quantum-gravity interaction [19]. This differs from collective tunneling of stellar matter through its degeneracy barrier, which can be far more suppressed [18]. We therefore consider finite-lifetime scenarios and a stable-baryon branch rather than assigning one disappearance date. The assumed channel must specify which bound and free nucleons decay.

Condition	Event	Physical description and calculation
Decay heating exceeds other sources	Decay-supported luminosity	Compare deposited decay energy with cooling and environmental power. This can occur long before a substantial fraction of the mass is lost. Escape fractions depend on decay products and the evolving column density.
Accumulated decays	Composition changes and mass loss	Track baryonic mass, nuclear daughter products, subsequent decays and reaction energy. Begin with coarse composition groups and refine only where they change pressure, retained heat or phase order.
Pressure support changes	Loss of degeneracy	Follow the transition from a degenerate remnant to matter whose structure depends on atomic and molecular forces. Recalculate the mass-radius relation; do not impose a transition mass from a different composition.
Columns become small	Decay products and thermal radiation escape	Separate transparency to energetic decay products from transparency at thermal wavelengths. Decreasing energy retention changes luminosity and temperature; an optically thin object need not radiate as a blackbody.
Binding becomes insufficient	End of the bound remnant	Allow fragmentation or dispersal if the material and energy balance require it. This is distinct from decay of every formerly associated baryon. Accretion, capture or collision can interrupt the isolated path at other stages.
Few baryons remain	Discrete survival and final decay	Replace a continuous mass variable with particle survival probabilities. Specify any stable products. If all original baryons have finite lifetimes, their last decay is a random event; an exponential mean never reaches zero at a finite time.
Stable-baryon alternative	No decay-imposed endpoint	Continue cooling, nuclear and environmental alternatives without assigning a proton-decay date. Any gravitational destruction channel requires its own assumptions and rate.

The degree of physical detail follows the quantity being predicted. Resolved atmospheres and interior transport determine the stellar track and cooling ages. At more uncertain stages, thermal balances, encounter distributions and conditional decay models are sufficient unless additional structure changes the result. Numerical errors must still be smaller than the signal of interest, especially for weak tidal heating. Successive descriptions will be compared where both apply, preserving mass, composition and stored energy.

8.4 A reference clock for the decay era

A useful analytic comparison assumes that every baryon has the same constant, independent mean lifetime τ_B, with no accretion or stable daughter baryons. Let t_d measure elapsed time from the start of this comparison. Then

$$\langle M_B(t_d)\rangle = M_{B,0}e^{-t_d/\tau_B}, \qquad L_{\rm heat} = f_{\rm dep}(t_d)\frac{\langle M_B\rangle c^2}{\tau_B}. \tag{1}$$

The second expression treats baryon mass as the available rest energy; a physical channel also accounts for daughter masses and nuclear binding. The deposited fraction $f_{\rm dep}$ is allowed to evolve. This is a reference mass and energy model, not a resolved calculation of the late atmosphere.

Expected initial baryonic mass lost	t_d/τ_B
50%	0.6931
90%	2.303
99%	4.605
99.99%	9.210

For N_0 independently decaying baryons the last-event distribution is

$$\Pr(t_{\rm last} \le t_d) = \left(1 - e^{-t_d/\tau_B}\right)^{N_0}, \qquad \langle t_{\rm last}\rangle = \tau_B H_{N_0}, \tag{2}$$

where $H_N = \sum_{k=1}^{N} 1/k$. For an initial baryonic mass $0.1M_\odot$, $N_0 \simeq 1.2 \times 10^{56}$. The mean last decay occurs near $129.7\,\tau_B$, with a central 90% interval from 128 to $132.1\,\tau_B$. This follows all tagged original baryons, including any dispersed material; it does not date the survival of a bound star. Correlated processes, composition-dependent lifetimes and stable products change this reference.

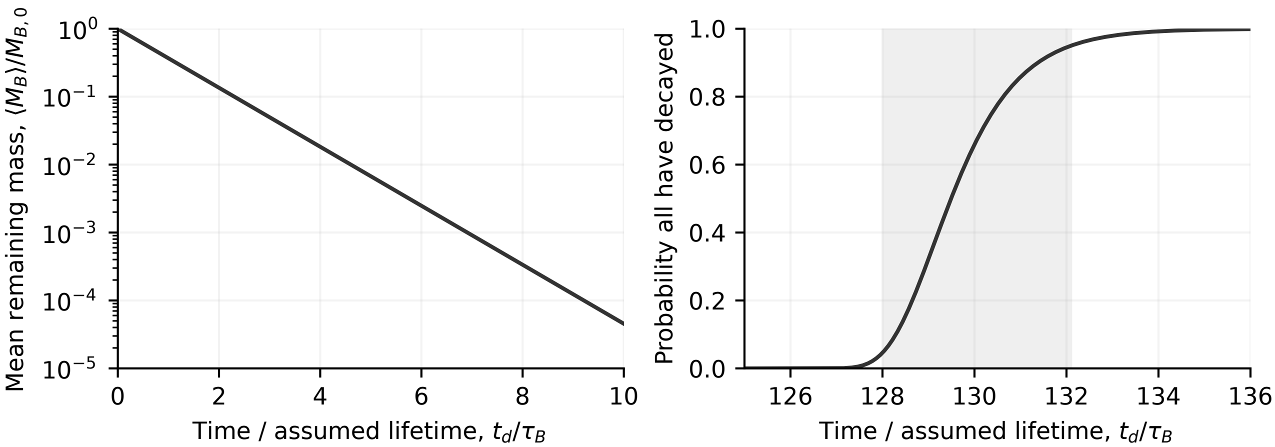

Figure 5: Conditional decay reference for an initially $0.1M_\odot$ baryon population with one independent lifetime. Left: expected remaining mass. Right: probability that all original baryons have decayed; shading marks the central 90% last-event interval. The two time ranges differ. These curves are analytic consequences of the stated assumptions, not an extrapolation of the computed Ember track.